

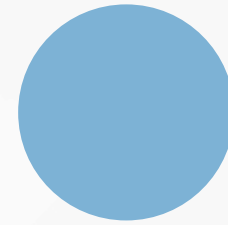
TELECO CUSTOMER CHURN -PREDICTION

IT 326- Data Mining

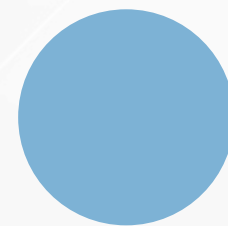
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PROBLEM DEFINITION



Customer churn is a major issue for telecom companies because losing subscribers directly lowers revenue and increases the cost of finding new customers. Churn happens when a customer stops using the service. By predicting churn, telecom providers can take early action, like offering personalized deals or improving service, to keep high-risk customers from leaving.



In our project we studied telecom customer data, and this data helped us identify the key factors and risks that can lead to customer churn.

DATASET

The dataset used in our project is the **Telco Customer Churn dataset(Kaggle)**. It includes customer demographics, account information, service subscriptions, contract type, and whether the customer churned.

It contains:

- 7,043 instances
- 21 features and (19 after preprocessing)

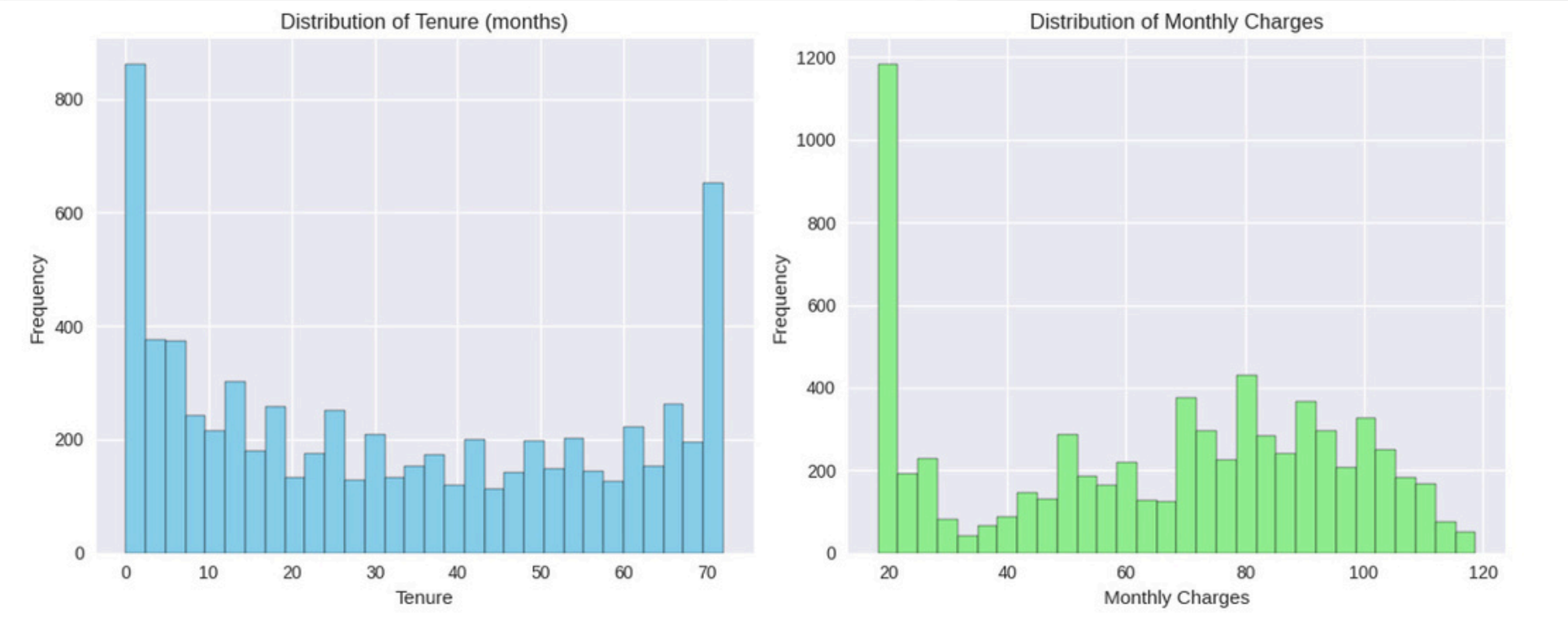
This dataset is **imbalanced**, there's more no than yes.

Data source:

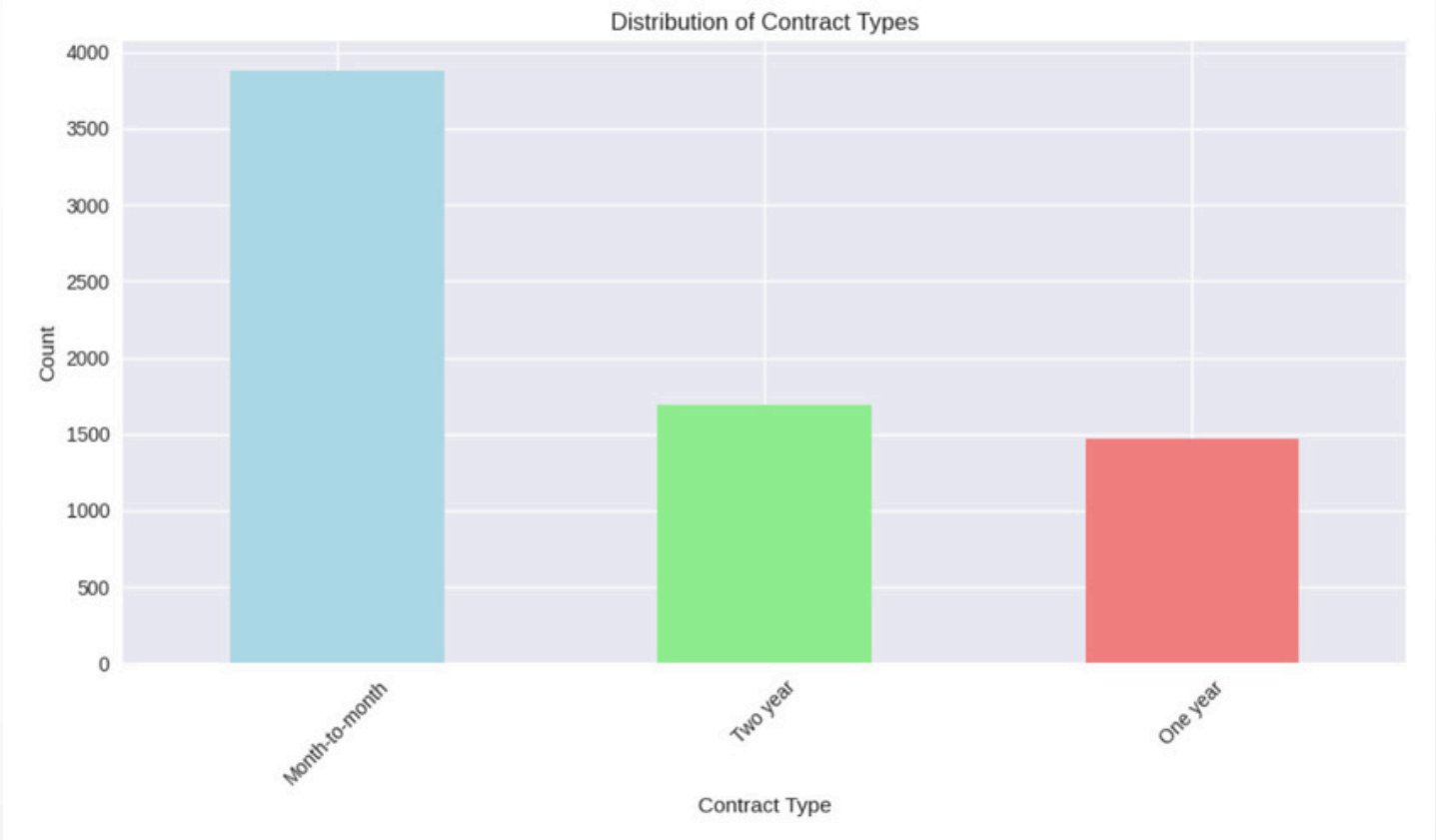
<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>



OUR DATA GRAPHS

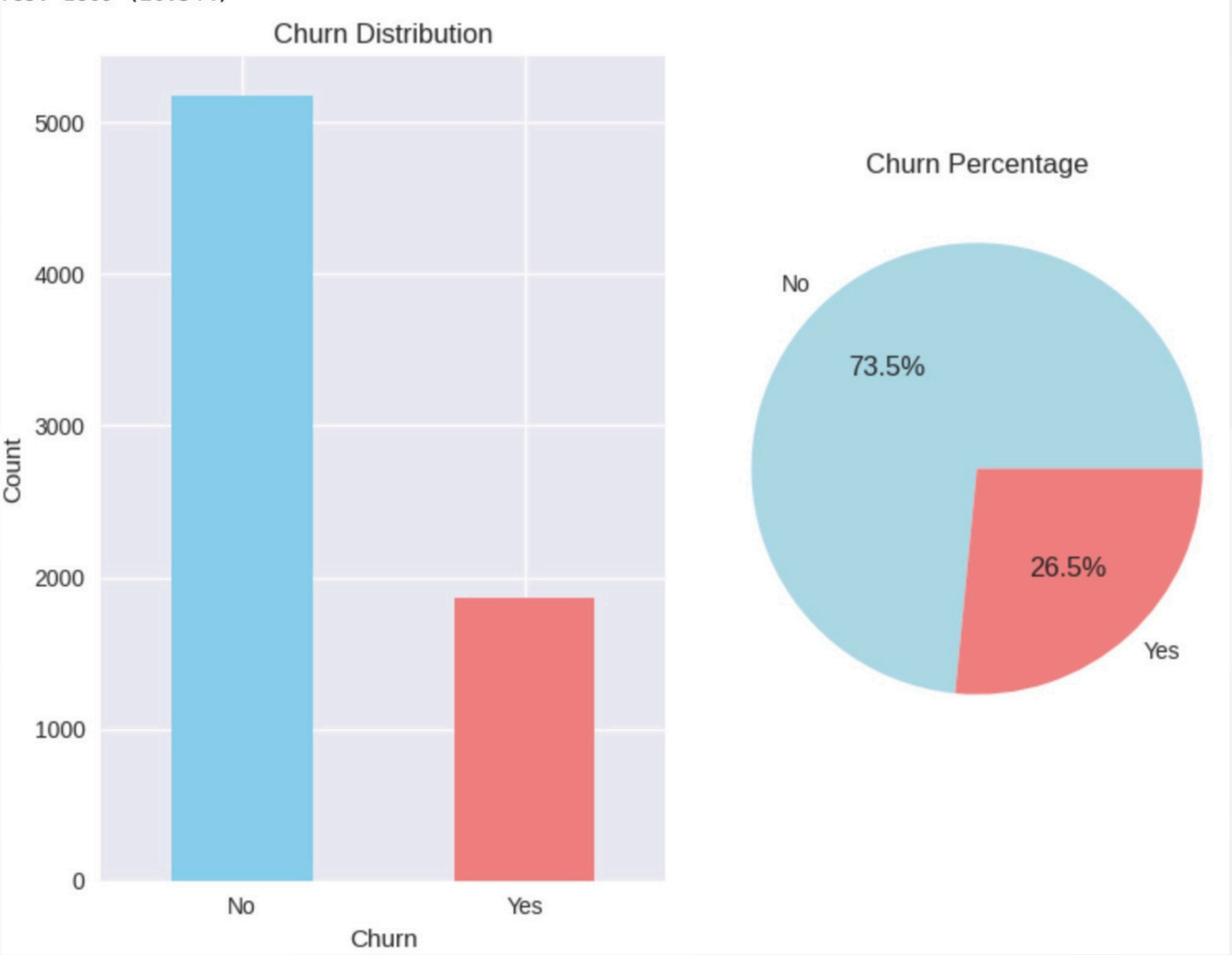


HISTOGRAM

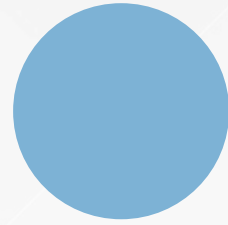


BAR PLOT

CHURN DISTRIBUTION

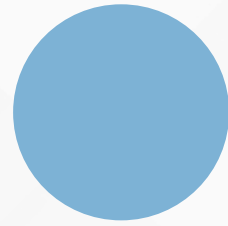


DATA CLEANING



We examined the dataset for non numeric and missing values especially in TotalCharges.

All invalid entries were fixed or converted to a valid format.



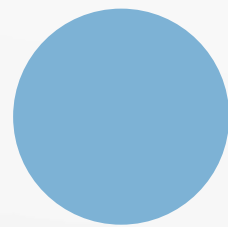
We verified potential outliers using:

- Box plots

- Five number summary

- IQR METHOD

- No harmful outliers were detected, so no rows were removed.



Duplicate rows were checked, and dataset consistency was validated before modeling.

DATA PREPROCESSING

To improve model performance and ensure consistent input quality, several preprocessing steps were applied to the dataset. These steps helped prepare the data for both classification and clustering tasks.

- Handling missing and inconsistent values
- Encoding categorical variables
- Normalization of numerical features (eg: tenure, charges)
- Outlier inspection and verification
- Final dataset restructuring

RAW DATASET

Sample of dataset:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	Churn
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	Yes

5 rows × 21 columns

NO INTERNET SERVICE

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	Churn
0	0	0	1	0	-1.277445	0	1	0	0	2	0	0	0
1	1	0	0	0	0.066327	1	0	0	2	0	2	0	0
2	1	0	0	0	-1.236724	1	0	0	2	2	0	0	1
3	1	0	0	0	0.514251	0	1	0	2	0	2	2	0
4	0	0	0	0	-1.236724	1	0	1	0	0	0	0	1

PREPROCESSED DATASET

DATA MINING TECHNIQUES

- 1- Classification
- 2- Clustering

CLASSIFICATION TECHNIQUE

Algorithm: Decision Tree Classifier (supervised)

Target: Predict Churn Yes/ NO



Why used?

- Interpretable
- Works well with mixed features
- Good baseline for churn prediction



Configurations:

- **Train/Test Splits:** 90/10, 80/20, 70/30
- **Criteria:** Gini index & information gain (Entropy)

OUR CLASSIFICATION RESULTS

All models achieve around 72–74% accuracy.
Entropy is slightly better than Gini in all splits.

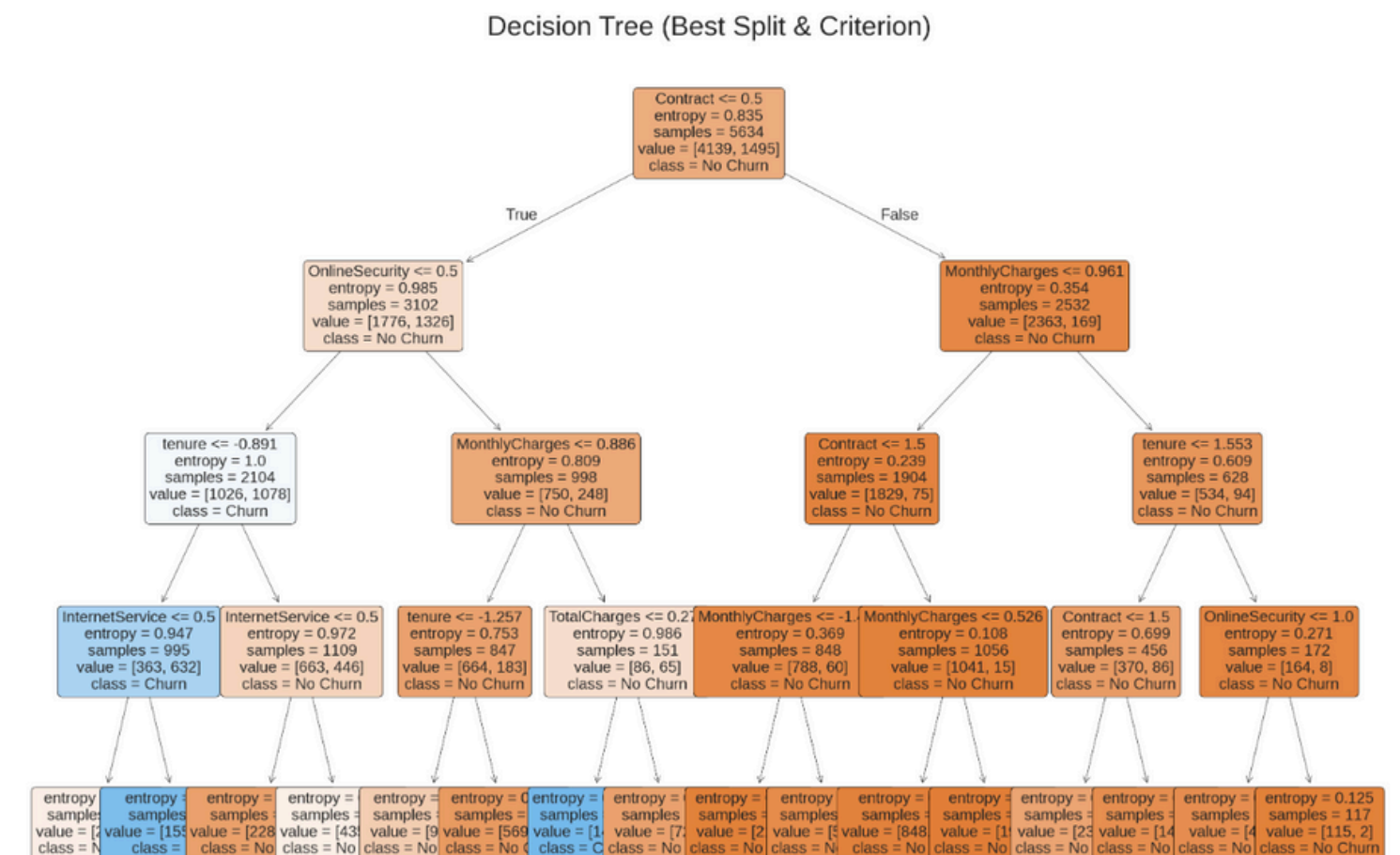
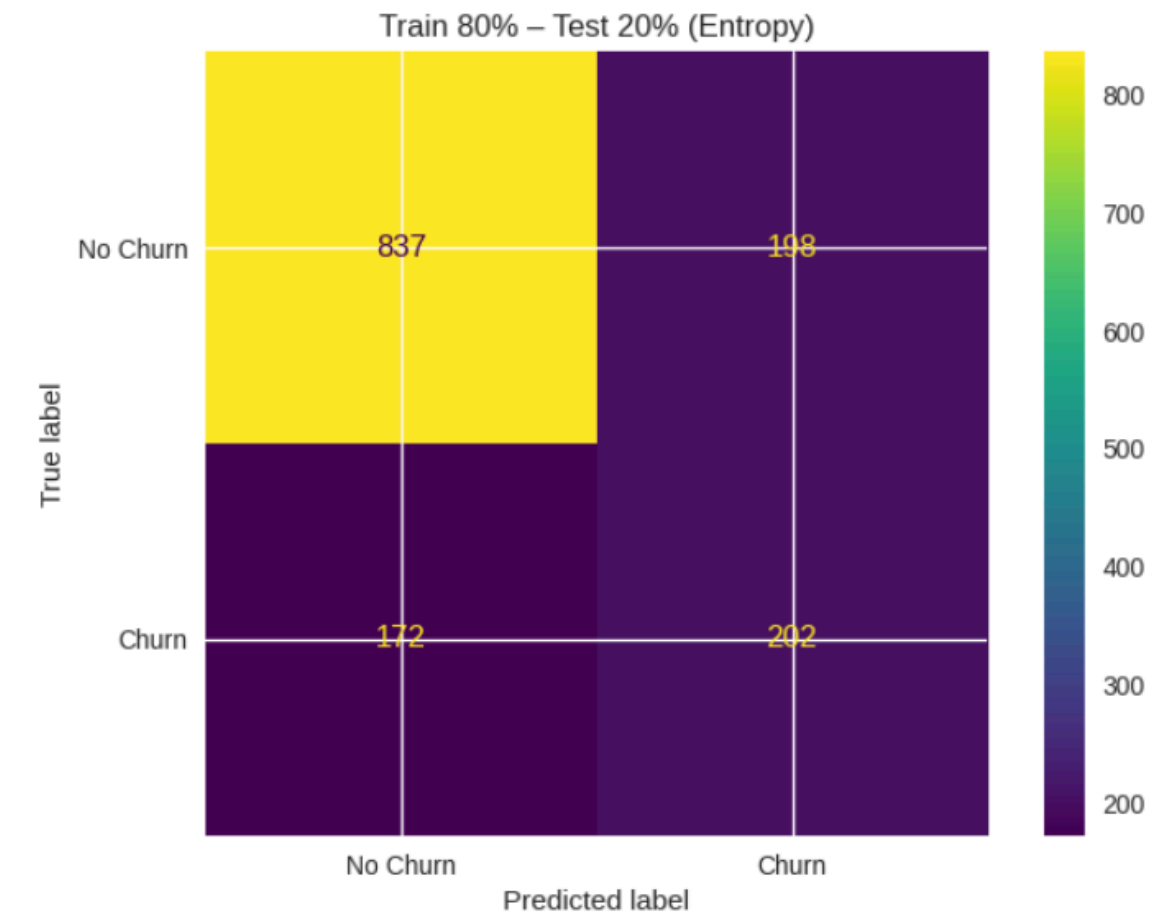
- **Best model:** Entropy with 80/20 split
- **Reason:** Higher accuracy and more stable performance across metrics

Train %	Test %	Gini Accuracy	Entropy Accuracy
90%	10%	0.7234	0.7291
80%	20%	0.7282	 0.7374
70%	30%	0.7312	0.7331

OUR CLASSIFICATION INSIGHTS

- The model predicts **non-churn** with high accuracy
- It struggles more with predicting churn (Yes), BUT it's still useful for early warning.

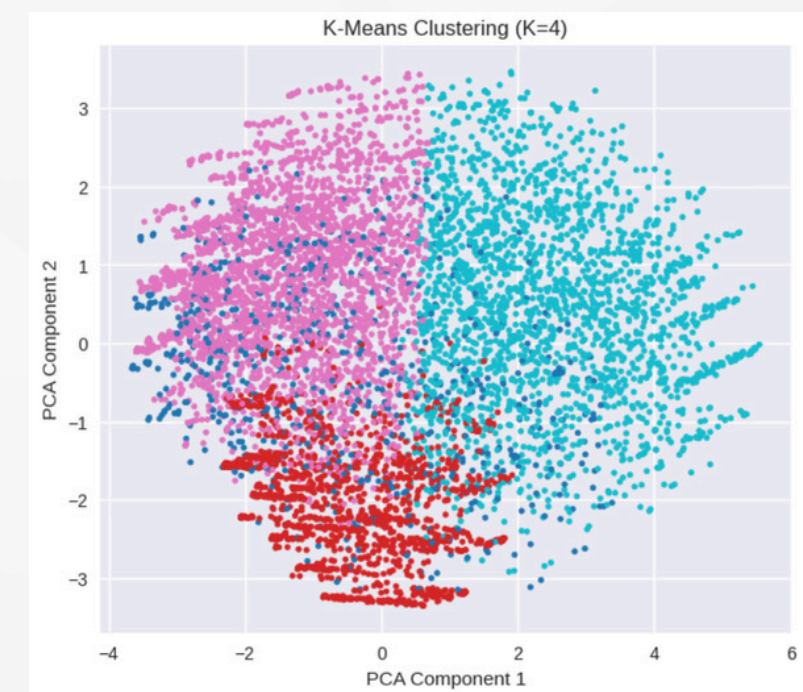
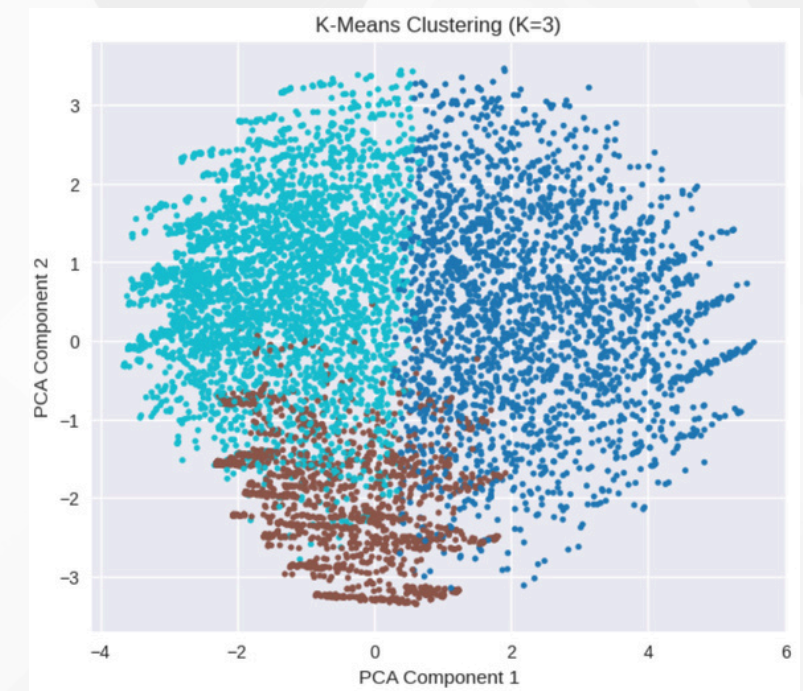
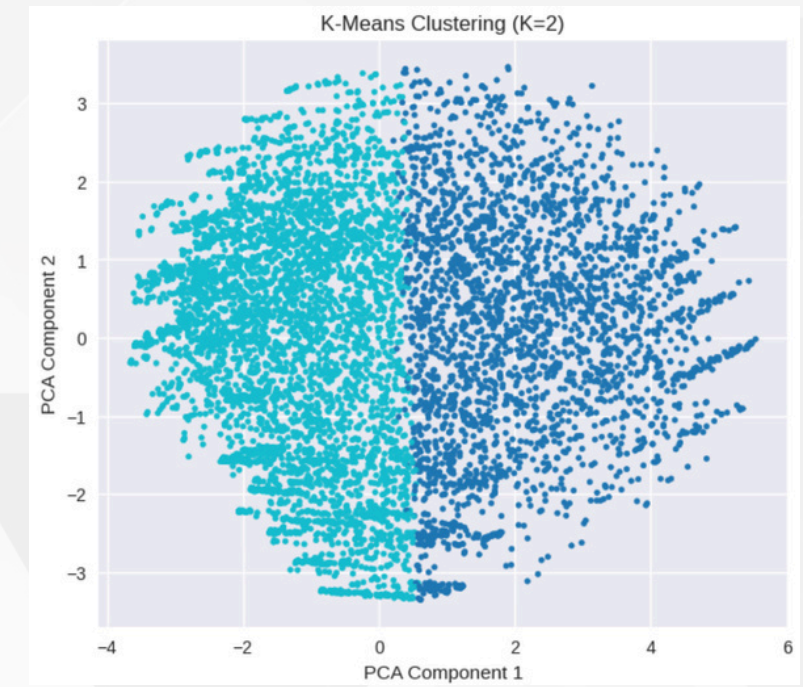
- Key features influencing the tree include **tenure**, **contract type**, and **monthly charges**.
- The tree shows that month-to-month contracts and higher charges are the **highest churn risk group**.



CLUSTER ANALYSIS

Algorithm: K-Means (unsupervised)

- Tested $K = 2, 3, 4$
- Evaluated with Silhouette Score + WCSS
- PCA visualization used



OUR CLUSTERING RESULTS

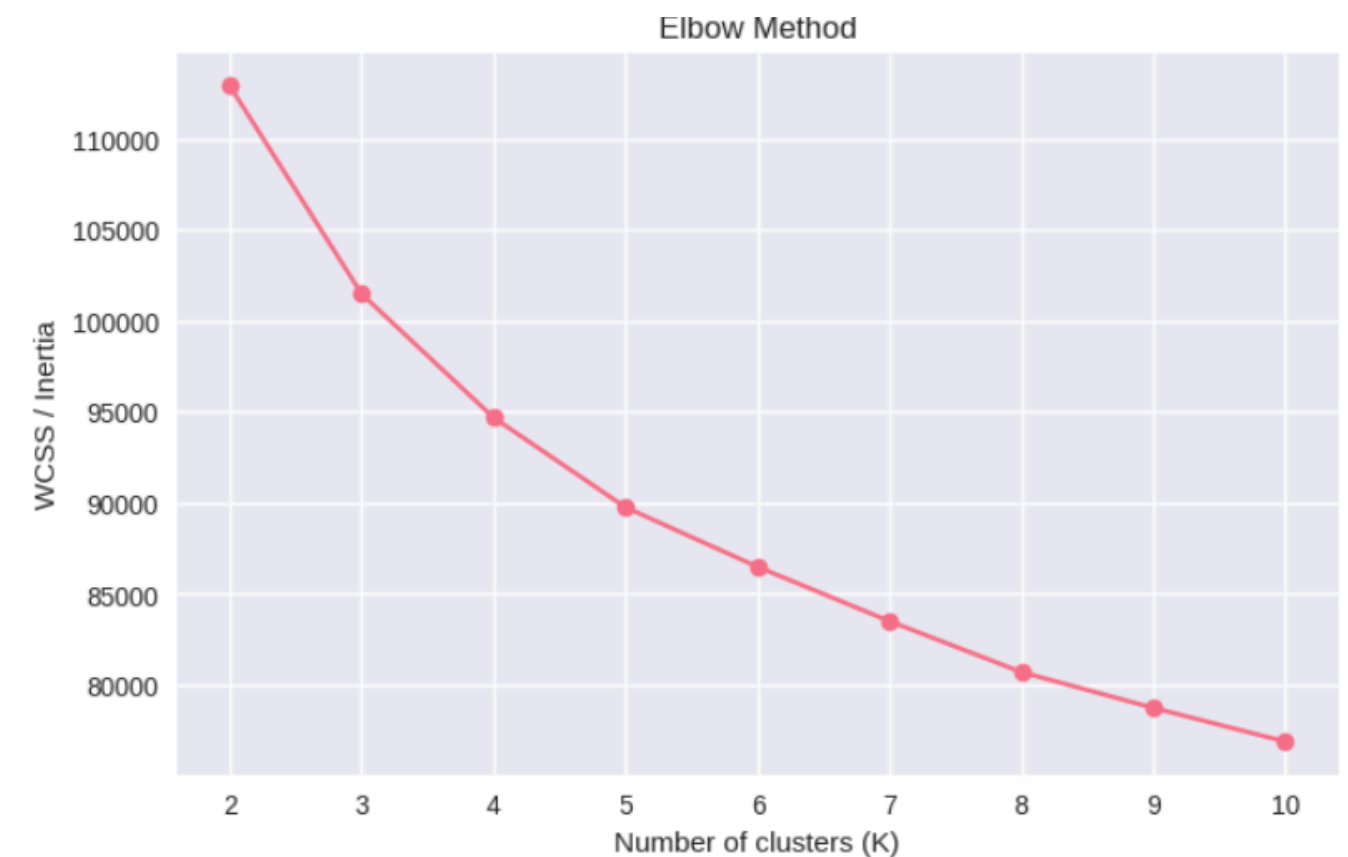
- **Optimal K = 2 (highest silhouette)**

(Clearer separation in PCA plot)

- Clusters overlap heavily → weak natural grouping
- WCSS decreases with higher K, but separation weakens
- Clustering not very effective for this type of dataset

	K	Silhouette	WCSS
0	2	0.157138	112918.372539
1	3	0.138061	101487.124098
2	4	0.155302	94693.531065

Results Table



Elbow Plot

FINAL RESULTS AND FINDINGS

CLASSIFICATION > CLUSTERING

- Decision Tree models achieved ~72–74% accuracy.
- Entropy with 80/20 split produced the best accuracy (73.74%).
- The most influential features for predicting churn were:
 - Tenure
 - Contract type
 - Monthly charges
- The classifier identifies churn risk reasonably well and supports early customer retention decisions.



COMPARISON WITH RELATED WORK

- The selected research paper also used a Telco Customer Churn dataset.
- Their best model (e.g., Random Forest / Gradient Boosting) achieved higher accuracy than our Decision Tree, but at the cost of higher complexity. Their accuracy score is 93.3% using XGBOOST but their decision tree accuracy 72.2% which is similar to ours.

- **Similarities:**

- Important features: contract type, tenure, monthly charges.
- Churners often have short tenure and month-to-month contracts.
- Our results support the same patterns, confirming that these features are strong churn predictors.



THANK YOU !